

A Literature Survey on Multimodal Route Optimization and Real-Time Traffic Prediction for GPS Tracking Systems

Ashish Sharma: Scientist, Regional Remote Sensing Centre (RRSC), Nagpur, India.

Dr. Reema Roychaudhary: Assistant Professor, Computer Engineering Department, St. Vincent Pallotti College of Engineering & Technology (SVP CET), Nagpur, India.

Kashish Joseph, Rochan Awasthi, Shreya Doye, Suraj Dhere: Students, Computer Engineering Department, St. Vincent Pallotti College of Engineering & Technology (SVP CET), Nagpur, India.

Abstract- This survey discusses the issues and techniques related to digital network mapping, shortest path analysis, and real time traffic management. This paper creates a basis to develop multimodal GPS tracking and navigation systems by synthesizing the research on multimodal metaheuristics, adaptive topology optimization, and deep-learning-based traffic forecasting. The survey has identified the current gaps in live multimodal data integration in addition to major strides that have been made in computational efficiency and accuracy.

Keywords—multimodal optimization, shortest path algorithms, real-time traffic prediction, GPS tracking, graph modeling.

I. INTRODUCTION

The pace of development of digital technology has greatly contributed to the complexity of transportation networks in cities. Effective management of urban traffic has become essential in alleviating congestion, as well as improving the commuter experience. This survey will deal with the fundamental goals of developing a system that will digitally map road networks, save the GPS location data and calculate the best routes by the application of a graph-based algorithm.

A. Motivation and Problem Statement : The complex effect of transfer times and conflicting requirements (time vs. cost) in a multimodal environment is often ignored by contemporary navigation systems. Moreover, static routing algorithms are often based on a failure to capture the dynamism of real-time traffic and thus poor estimates of travel time. The aim is to test

methodologies that facilitate real-time navigation and effective route planning by proper shortest-path identification.

B. Scope of the Survey Four key research areas are explored in this survey: multimodal metaheuristics to multi-criteria path optimization, extensions of adaptive topologies to rail transit, real-time spatio-temporal traffic prediction, and access-node routing to integrate heterogeneous networks.

II. LITERATURE SURVEY

A. Modeling Frameworks and Multimodal Resilience

The paper by Akinyemi et al. is a review of the modeling solutions employed in multimodal transportation systems. Their study highlights an important concept of network resilience, which is the ability of the system to continue to be operationally functional in the face of either internal or external stress. The paper divides transportation models into two paradigms: user-optimal behavior, where people solve their own time in personal transportation and system-optimal behavior, which aims at the minimization of the total cost in the entire system. Through pattern analysis of integrated mobility, the authors emphasize that multimodal infrastructures need to reconcile supply side constraints and evolving commuter demands. This background review contextualizes the need to have adaptive navigation systems that have the capability of responding to network disruptions in real-time.

B. Multi-Criteria Vehicle Route Optimization

The use of classical algorithms in the modern vehicle routing is discussed in the works on multi-criteria optimization in Dijkstra algorithm. The study resolves the constraints of single-objective pathfinding by proposing a framework that considers the three factors, travel time, distance, and monetary cost, as weighted. The strategy would be to convert these different criteria into one goal action by means of weighted averaging operators. The notable result of this study is that the optimization of vehicle route can only be done with individualized cost functions because of the cost difference on different fuels and user priorities. This research offers an essential criterion to use in computing the best routes to follow when transporting things, which is a fundamental output of GPS trackers.

C. Complex Network Theory in Urban Transit

Cui et al. implement the Complex Network Theory to the optimization of urban public transport networks. The authors employ graph-theoretic measures, such as degree centrality and betweenness centrality to determine critical nodes and bottlenecks in the city transit map. Through the modeling of the transport system as an intricate network, the study finds the network robustness to be strongly linked to the connection of certain hub stations. According to their results, these nodes are the main ones that should be reinforced as a part of the optimization strategies that will improve the overall efficiency of transit. The approach is critical in the process of digitalizing road networks because it enables the determination of busy intersections that determine the efficiency of routing.

D. Temporal Networks from GPS Data

The article by Kato et al. explores how time-stamped GPS data can be used to construct temporal networks, in particular in the case of

Kyoto tourism. Temporal networks also model the time-dependent quality of routes and city congestions, unlike other conventional networks that are not time-dependent. The authors have worked out the methodology to generate historical GPS records to identify trends of tourist movement and seasonal traffic changes. This paper shows that the temporal network-based route optimization provides much better travel-time estimates than a static model because it considers the time-varying capacity of road segments. This methodology can be directly applied to future developments in AI-based congestion modeling and route prediction.

E. Multi-Criteria Shortest Path via Morphological Filters

Amar et al. offer a new metaheuristic which is called Optimization by Morphological Filters (OMF-MT) to solve the Multi-Criteria Shortest Path (MCSP) problem in multimodal networks. Image processing was inspired by the methodology, which uses an erosion transformation to search through the space, attempting to find local optima around a structuring element or morphological filter. The system uses a series of parallel filters to guarantee such vigorous search and add a diversification technique to avoid blockage at local optima. In a study of four modes of transport (metro, tram, bus, and taxi) and three criteria (distance, time, and cost), OMF-MT showed a huge benefit in the computation time as compared to Genetic Algorithms (GA) and NSGA-II. The algorithm will form the core of calculating optimal multimodal paths in our proposed system because it is capable of giving good quality solutions at low computational costs.

F. Spatio-Temporal Traffic Prediction Framework

In order to make the jump towards traffic-sensitive navigation, Kumar et al. propose a framework of real-time traffic prediction that combines Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks. The approach uses CNNs to learn spatial features of traffic sensors and GPS locations, and LSTMs to learn the intricate time dependences of city traffic. Their CNN-LSTM model attained an 89 percent prediction rate regarding the issue of traffic

congestion, which is way higher than the traditional ARIMA and Random Forest model. Besides, this research applies Deep Reinforcement Learning (DRL) to adaptive traffic signal control, which decreased the delays by 30% on average. This project will adopt this framework to meet the scope of the project in the future, which is the prediction of routes and calculation of travel-time using AI.

G. Adaptive Topology for Urban Rail Accuracy

The inability to take into consideration the transfer times in interchange stations usually impairs the accuracy of the urban rail navigation. Fang et al. overcome this by offering the Adaptive Topology Extension Road Network (ATEN) structure. The methodology classifies transfer stations according to their physical structure and provides virtual transfer stations to clearly distinguish between transactions paths of pedestrians and train movement paths. The ATEN structure enables a regular optimized Dijkstra algorithm to query optimal shortest paths without the use of complicated real-time operations on transfer nodes by considering transfer time and travel time as the same edge weight. This is a structural approach that will be used to make sure that deliverable of the shortest route accurate of our digital road network mapping.

H. Access-Node Routing for Hybrid Integration

Tischner solves the problem of integrating high volume transit schedules and road systems with the Access-Node Routing (ANR) concept. In this methodology, the multimodal routing is made possible by choosing particular access nodes in the transit network where a user can leave the road based travel. Connection Scan Algorithm (CSA) is used in the backend implementation to use time-dependent transit data, and Tischner discovered it to be up to 1,000 times more efficient than Dijkstra-based implementations on large-scale schedules. On the road segments, the goal-directed efficiency is ensured by using a modified version of the ALT algorithm (Landmarks and Triangle Inequality). This architectural approach will be used in the backend of our system in order to support the coordination of the dissimilar data

formats (OSM road data and GTFS transit feeds) and a low query latency.

III. BASE PAPER SELECTION AND IMPLEMENTATION STRATEGY

This paper chooses Amar et al. (OMF-MT) [1], Kumar et al. (CNN-LSTM) [2], Fang et al. (ATEN) [3], and Tischner (ANR/CSA) [4] as the four initial papers to implement after a rigorous literature review.

These particular works are chosen based on the fact that they are directly applicable to the modules of our project:

1. **Network Mapping:** We will implement the ATEN structure [3] to accurately model transit transfers within our digital map.
2. **GPS Tracking & Optimization:** OMF-MT [1] will be the primary algorithm for calculating multi-criteria shortest paths (time, cost, distance).
3. **Real-Time Data:** The CNN-LSTM framework [2] will be implemented to provide AI-based route prediction and traffic-aware optimization.
4. **Backend Efficiency:** ANR [4] will be used to manage the integration of high-volume transit timetables with sparse road graphs.

IV. COMPARATIVE ANALYSIS

The synthesis of the eight works surveyed indicates a profound methodological shift, towards more advanced AI-based and metaheuristic models, as opposed to the traditional graph-based extensions. In this section, these approaches are compared in terms of their computational processes, use of data and how they can be applied in real world navigation.

A. Methodological Comparison

The methodologies can be broadly categorized into three tiers:

- **Classical and Theoretical Frameworks:** Such works include Akinyemi et al [5]. and Cui et al. [7] are aimed at setting the theoretical strength and strength of networks. These give the required metrics of graph health, but are not usually as agile as needed by modern GPS tracking.
- **Algorithmic Extensions:** The multi-criteria Dijkstra [6] and ATEN [3] algorithms are extensions to classical pathfinding which introduce complexity layers, like virtual stations and weighted multi-objective functions. They are very precise, but need extensive processing of network topologies.
- **Advanced Metaheuristics and AI:** The three main base papers are OMF-MT [1] CNN-LSTM [2] and ANR/CSA [4] which reflect the state of the art. The models put a priority on computational speed and predictive ability, thus they are applicable in a live system where the traffic conditions are subject to quick change.

B. Data Usage and Optimization Trends

It can be seen that the trend is away toward the use of data-driven models of temporal networks as opposed to static graph modeling. Kato et al. [8] show how historical GPS logs can be used to inform the route availability and Kumar et al. [2] go a step further to utilize real-time data of IoT sensors. One of the major differences is the treatment of the cost of a path by these papers: newer methods (OMF-MT) [1] consider cost as a dynamic multidimensional vector expressed as and older models tend to reduce cost to a single weighted scalar.

C. Summary Table of Methodologies

Table I provides a structured comparison of the core methodologies analyzed in this survey.

TABLE I. COMPARATIVE SUMMARY OF ROUTING AND PREDICTION FRAMEWORKS

Methodology	Data Type	Primary Mechanism	Key Strength	Compute Complexity
Complete	Static	Centralized	Identify	Moderate

Network	Network Map	Key Metrics	Key Bottlenecks	Efficiency
Temporal GPS	Historical GPS Log	Time Stamped Edges	Capture seasonality	High
OMF-MT	Multimodal Graphs	Morphological Filters	Rapid Global Convergence	Low
CNN-LSTM	Real-Time IOT Data	Spatio Temporal DL	89% forecast accuracy	High
ATEN	Transit Topology	Virtual Stations	Corrects transfer errors	Moderate
ANR/CSA	GTFS/OSM Data	Access Node Routing	Scales to country level	High

One of the trends is the shift towards local search mechanisms (OMF) and sequence learning (LSTM) to manage non-linear high-dimensional traffic patterns. Nevertheless, there is still a gap in the seamless connectivity of these deep learning models to large-scale multimodal graphs, with most high-accuracy predictive models being tested on particular metropolitan datasets as opposed to being integrated, cross-modal networks.

D. Identified Research Gaps

In spite of these innovations, there is still a gaping hole in the seamless combination of real-time predictive analytics with multimodal transfer logic. Although Kumar et al [2]. provide high-accuracy traffic predictions, their model is mostly used in single-mode traffic flow. Multi-modal designs on the other hand such as Amar et al [1]. and Tischner [4] have very good routing logic but frequently use average or constant values of travel time instead of the live, predicted congestion values. This gap is critical in meeting the project goal of efficient route planning in a live environment.

V. DISCUSSION

The results of the eight researched papers would give a harmonious technical roadmap on how to develop the V-NEURON and its essential goals. The synthesis implies that an architecturally accurate and computationally agile GPS tracking system should be truly adaptive.

- **Topological Accuracy and Resilience:** The ATEN structure [3] is directly related to the deliverable of the project in the form of the digital road/network map. The system would be able to provide high fidelity navigation in a city such as Nagpur by introducing virtual stations to separate pedestrian and train routes. Such structural accuracy is further supported by Complex Network Theory [7], which determines critical hub nodes that determine network robustness and efficiency. In addition, the theoretical model of network resilience [5] is such that it is able to operate under the dynamic pressure of the urban traffic.
- **Multi-Criteria Optimization:** The task of shortest route analysis demands an escape of single-metric pathfinding. OMF-MT algorithm [1] has the required speed to support real-time requests through parallel morphological filters in order to locate global optima. The system can be used to provide custom advice on the various types of vehicles when used with the personalized cost functions in multi-criteria vehicle routing, which is weighted by fuel, time, and distance [6].
- **Data-Driven Intelligence:** The CNN-LSTM framework [2] integration satisfies future of the AI-based route prediction. Extraction of spatial patterns through CNNs, and temporal dependencies through LSTMs, provides the system with the predictive capability (89% accuracy) to predict congestion in advance to manage it. This is complemented by the fact that time networks are made using GPS data [8]

which enables the system to use the past logs to record time-dependent route availability.

- **Backend Scaling:** In order to handle the "digital database of GPS tracking" there is Access-Node Routing (ANR) and Connection Scan Algorithm (CSA) [4] as a method of dividing high-volume transit schedules and sparse road maps. This division minimizes the memory overhead, which is a major concern with cloud-based navigation services.

V. CONCLUSION & FUTURE WORK

This survey has analyzed a diverse range of methodologies for multimodal shortest path optimization and real-time traffic management. Key insights include:

- The computational efficiency of **Morphological Filters** [1] for multi-criteria problems.
- The necessity of **Adaptive Topologies** [3] to eliminate search errors at transit transfer points.
- The predictive accuracy of **Hybrid Deep Learning (CNN-LSTM)** [2] for urban traffic flow.
- The scalability of **Access-Node Routing** [4] for integrating disparate network data.
- The identification of network bottlenecks through **Complex Network Theory** and the importance of **Temporal GPS logging** for accurate historical mapping.

The next steps in future work will be to refine these algorithms to ensure that they can support dynamic networks in which live environmental data, including weather and emergency service notifications, can play a major role in route choice. The following stages of the development will also address the opportunity to incorporate IoT-based vehicle tracking systems and apply the concept of Machine Learning to anticipate the preferences of individual travelers and make the system flexible to the changing needs of urban planning in smart cities.

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